

LIDA Composting System Software Testing Document

This document defines the testing plan, approach, environment, and structure for verifying the functionality and performance of the LIDA Composting System. It ensures that all system components including the 3D model, mobile application, and sensor simulation meet functional requirements and operate correctly when integrated.

The primary goal of this document is to:

- Establish a consistent testing framework for all team members.
- Record test cases, results, and evidence for verification and validation.
- Identify and resolve any system-level defects before final deployment.

Document Title	LIDA Composting System – Software Testing Document
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Version	Draft 1.0
Date Created	
Approved By	

Test Environment

Category	Description
Hardware	Laptop (i5/i7 processor, 8 GB RAM minimum) Mobile device (Android 10+ or iOS 14+)
Software	SolidWorks 2021 / SketchUp Pro, Flutter SDK, Visual Studio Code, Android Studio
Test Data	Mock sensor data (temperature, gas, moisture) using local JSON files or MQTT broker
Network	Local Wi-Fi connection / simulated offline testing
Testing Tools	SolidWorks Motion Manager, Flutter Hot Reload, Android Emulator, Screenshot Tools
Testing Standards	IEEE 829 Software Test Documentation Standard / Agile Sprint Testing Cycle

Testing Approach

The system integrates several components — the 3D SolidWorks model, mobile application interface, sensor simulations (temperature, gas, and moisture), and animation logic. Testing will confirm that these components work as intended both independently and together.

Testing will be **manual and black-box based**, focusing on user-observable results rather than code-level outputs. Each test will describe what component is being validated, why it's important, expected outcomes, and how results will be documented.

The testing process will follow a bottom-up approach: > Unit → Integration → System → Usability.

Each stage will inform the next: unit tests will verify individual module correctness, integration tests will validate communication between modules, system testing will verify end-to-end performance, and usability testing will evaluate user experience.

- Testing Levels:
 1. Unit Testing: Verification of individual modules (3D model components, animations, sensor logic).
 2. Integration Testing: Validation of interactions between modules (e.g., 3D models vs mobile app).
 3. System Testing: Full-system testing of performance, data flow, and UI behaviour.
 4. Manual Functional & Usability Testing: Evaluation of how intuitive and reliable the system appears to users.
- Test Execution:
 - Each team member executes their own unit tests.
 - Thenura consolidates integration testing and system testing results.
 - Screenshots and logs will be attached as evidence.

Earlier test phases will ensure smaller issues are corrected before larger-scale testing begins.

Roles and Responsibilities

Role	Name	Responsibility
Scrum Master / Lead Tester	Thenura Kuruppuarachchi	Coordinate testing activities and compile final test document.
Animation Tester	Haritha Perera	Test motion logic and animations for paddles and condensation.
Pipeline & Sensor Tester	Senupama Deshapriya	Validate pipeline flow, drainage, and sensor data accuracy.
UI and Documentation Tester	Chrishelle Anthoneyz	Verify UI data display, app export, and prepare evidence screenshots.

Acceptance Criteria

Category	Metric	Target
Performance	Model render time	< 2 seconds on mid-tier device
	Animation frame rate	≥ 24 FPS
	Sensor value latency	≤ 500 ms
Stability	Crash-free runtime	≥ 30 mins continuous
	Memory usage	Below device threshold
Accuracy	Sensor display vs mock data	100% match
Usability (Week 11)	Task success	≥ 80%
	Median user rating	≥ 4/5 for ease and clarity

Detailed Test Structure

1. Unit Testing

Unit testing focuses on validating the smallest components of the system independently. These tests will be manually executed using mock data and simulated animation logic.

Test ID	Module	Objective	Input/Action	Expected Output	Actual Output	Status	Evidence
UT-01	Paddle Animation	Verify bidirectional paddle motion	Run motion study	Paddles rotate smoothly			
UT-02	Drainage Pipe	Validate water flow	Play animation	Moisture animation flows consistently			
UT-03	Sensor Label Display	Confirm visibility of labels	Enable label view	Temperature, Gas, Moisture labels appear			
UT-04	GLB Export	Validate 3D export	Export GLB and import in app	Model opens properly in Flutter app			
UT-05	Colour Mapping	Verify per-part material persistence	Export GLB	All parts retain assigned colours			
UT-06	Animation Controller	Test play/pause functionality	Tap play → pause	Smooth toggling, no lag			

Each module will be tested using isolated mock data prior to integration testing. Expected results will confirm proper component behaviour before moving to cross-module testing.

2. Integration Testing

Integration testing ensures that modules communicate effectively. Tests will examine how the 3D model interacts with the mobile app and how sensor data is rendered on-screen.

Test ID	Integrated Modules	Objective	Expected Outcome	Actual	Status	Evidence
IT-01	3D Model + App Interface	Validate GLB rendering	Model loads without distortion			
IT-02	Sensor Panel + 3D Model	Check data sync on highlight	Selected sensor highlights on model			
IT-03	Sensor Data + Dashboard	Validate update timing	Values refresh within 5 seconds			
IT-04	Animation + UI	Test concurrent playback	Animation continues during UI input			

Planned Analysis: Integration will follow successful unit tests and confirm data consistency between modules. Any communication delays or rendering issues will be logged in the bug tracker.

3. System Testing (End-to-End)

System testing validates the full workflow — from app launch to animation completion. This ensures that the overall performance aligns with functional requirements.

Test ID	Scenario	Objective	Expected Behaviour	Actual	Status	Evidence
ST-01	Full Run	Test animation, sensor, and UI combined	Runs smoothly without lag or data loss			
ST-02	App Performance	Evaluate app loading and response time	Launch & load < 5 seconds			
ST-03	Data Consistency	Compare live vs mock sensor data	All values consistent			
ST-04	Offline Mode	Test network loss handling	Offline banner, cached values shown			
ST-05	User Interaction	Verify tab navigation	Smooth tab transitions, no freezing			

4. Manual & Black-Box Testing

Manual testing will simulate real-world use without code inspection. Both positive and negative cases will be included.

ID	Scenario	Action	Expected	Result
BB-01	App Offline	Disable Wi-Fi	Offline mode message, no crash	
BB-02	Sensor Update Delay	Slow network simulation	Values refresh < 1 sec delay	
BB-03	Invalid Sensor Data	Inject invalid data	Error message displayed safely	
BB-04	Highlight Function	Tap T1	Camera focuses on T1 sensor	

Usability Testing Plan (Week 11)

Participants: 5–8 general users unfamiliar with LIDA system

Devices: Android (A54, Pixel 6), iPhone 13

Each participant will complete structured tasks and rate them on a 1–5 scale.

User Tasks

1. Open the LIDA app and navigate to the dashboard.
2. Locate temperature, moisture, and gas sensors.
3. Play and pause the composting animation.
4. Interpret phase colours and sensor values.
5. Identify Active, Curing, and Storage phases on the model.

Survey Questions (5-point Likert)

1. How easy was it to navigate to the **3D tab** and understand how to begin interacting with the model?
2. Were the **buttons and controls** (e.g., Open Lid 1, Open Lid 2, Open Drawer) responsive and easy to use?
3. Was the **sensor data panel** (Temperature, Humidity, pH, CO₂) clearly visible and easy to interpret?
4. Did the **colour-coded chambers** help you identify the different composting phases (Active, Curing, Storage)?
5. How realistic and visually clear did the **3D model** appear (e.g., structure, layers, pipes, sensors)?
6. Could you easily **locate temperature, moisture, and gas sensors** on the model?
7. Was the **composting animation** (lid and drawer movement) smooth and easy to follow?
8. Were the **Play/Pause or control buttons** intuitive and functioning as expected?
9. Overall, how would you rate your **understanding of the composting process** after using the 3D tab?
10. What **improvements or additional features** would you recommend for enhancing the 3D model experience?

Analysis Plan

- Calculate **task completion %**, **average time**, and **median score per question**.
- Identify top **3 usability issues**.
- Log issues in the **Bug log** and assign owners.

Bug Log (Current & Future Tracking)

ID	Description	Environment	Steps	Expected	Actual	Severity	Owner	Status	Evidence
BUG-001	Paddles share same material after GLB export	Android 13	Import GLB	Retain unique colour per part	One uniform colour	High	Thenura	In Progress	Before/After screenshots
BUG-002	Sensor label delay	iOS 17	Tap T1	Value < 500ms	1.2s delay	Medium	Senupama	Open	Performance log
BUG-003	FPS drop during animation	Android 14	Switch tabs while playing	FPS $\geq$ 24	Drops to 18	Medium	Haritha	Ongoing	GPU profiler
BUG-004	Text overlap in dashboard	Android	Resize window	Auto wrap text	Overlapping text	Low	Chrishelle	Fixed	Screenshot
BUG-005	Temperature tab freeze	Android 13	Reopen app from background	Tab loads normally	App stalls for 3s	Medium	Thenura	Testing fix	Log file

Traceability Matrix

Requirement ID	Description	Test Cases	Evidence
RQ-01	Model must retain per-part materials	UT-05, IT-01	Screenshots, commits
RQ-02	Sensor highlight links to 3D model	IT-02, ST-01	Video clip
RQ-03	System runs under 2s load time	ST-02	Performance log
RQ-04	Offline mode displays banner	ST-04, BB-01	App screenshot

RQ-05	Usability and navigation feedback	Usability tasks 1–4	Survey summary
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Test Execution Schedule

Milestone	Assigned To	Date
GLB material verification (Unit)	Thenura	Oct 13
Sensor latency fix & retest	Senupama	Oct 16
Animation FPS test	Haritha	Oct 18
Usability pilot run	Chrishelle	Oct 20
Usability data analysis	Chrishelle	Oct 25
Full system retest & evidence	All	Oct 28–31

Reviewed By	Signature	Date
Supervisor (Mr. Naveed Ali)		
Scrum Master (Thenura Kuruppuarachchi)		